Assignment B *Climactor V2 visualizations*

01 Geographic Emissions Distribution — Slide 2

02 Population vs. Per-Capita Emissions — Slide 3

03 Inventories vs. Climate Plans & Targets — Slide 4

04 Initiatives vs. Quantified Targets — Slide 5

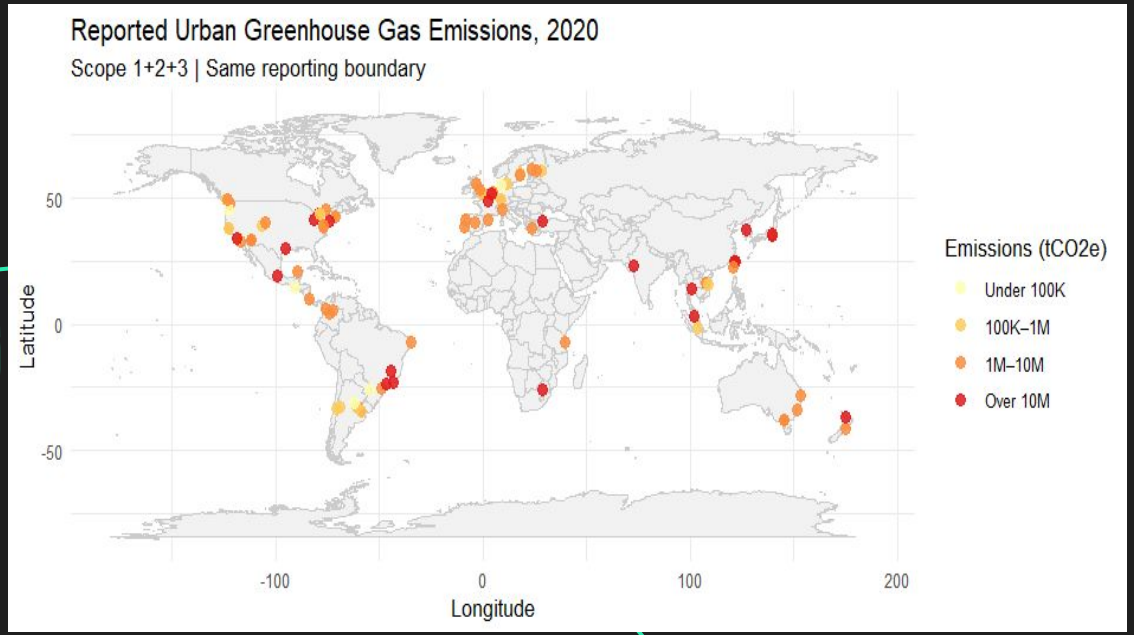
05 Total Emissions by Sector — Slide 6

06 Geography of Climate Planning History — Slide 7

Confidential

1. Where Are the Highest-Emitting Jurisdictions?

Reported greenhouse gas emissions were mapped to display geographic patterns and identify major emitters. To standardize, I selected 2020 inventories reported in tCO_2e with Scope 1+2+3 and a consistent reporting boundary, then used color categories to distinguish emissions levels. However, reporting coverage is incomplete, and optimizing for standardization of reporting year necessarily omitted many records. Additionally, possible differences in initial inventory methods remain even after standardizing scopes and boundaries.



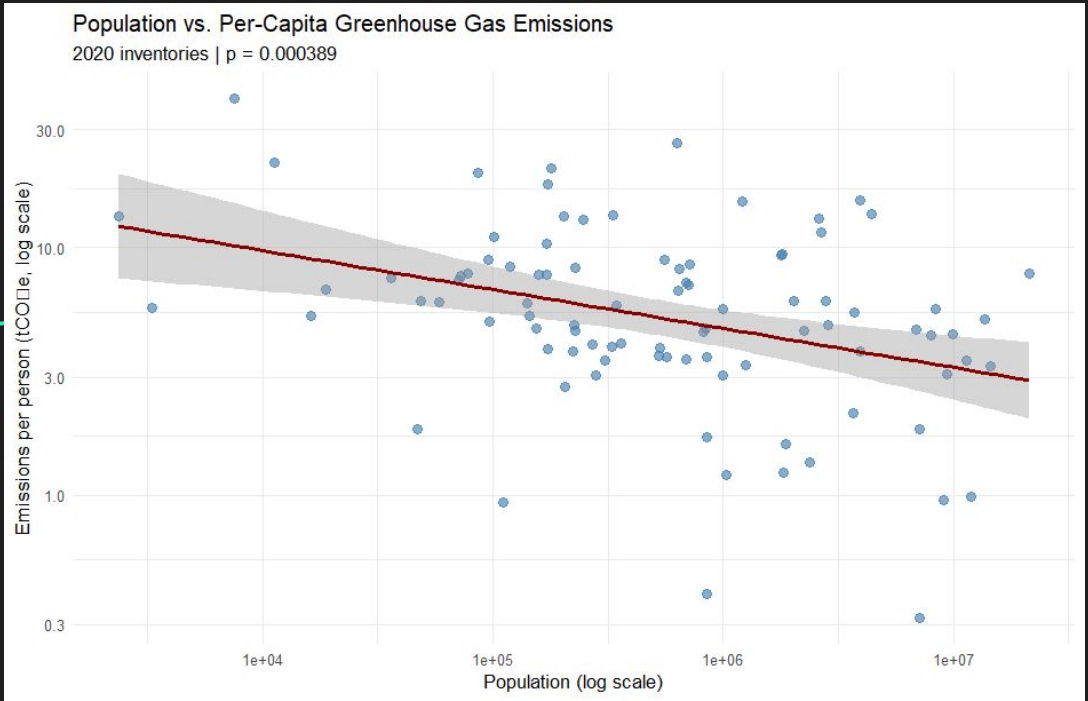
Why: Identify major reported emitters and reveal gaps in comparable reporting

Data quality: Restricted to comparable 2020 inventories; duplicates prioritized by source

Updates: Add new inventories and standardize scopes, boundaries, and units

2. Does Population Size Relate to Per-Capita Emissions?

Jurisdiction population is compared with emissions per capita to distinguish total environmental impact from emissions intensity. Logarithmic axes make it easier to examine relationships across jurisdictions that differ greatly in size. The relationship is highly significant.



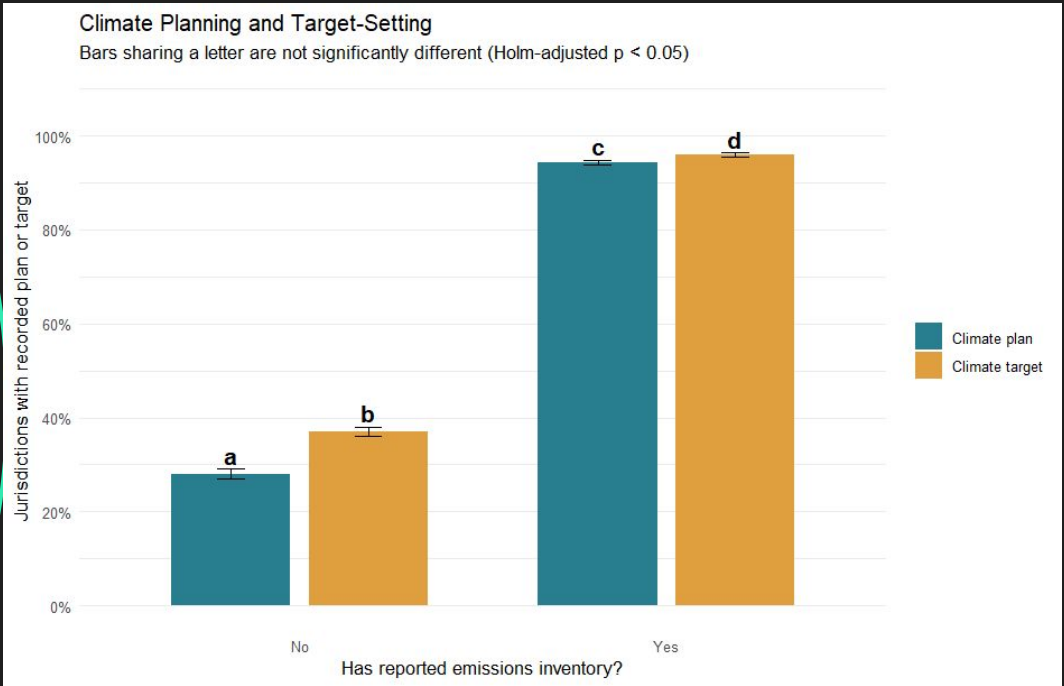
Why: Explore whether population size relates to emissions intensity, beyond total emissions alone

Data quality: Missing population data and extreme values may skew comparisons

Updates: Refresh population estimates to correspond with year and investigate outliers

3. How Does Climate Planning Vary With Emissions Reporting?

Prevalence of climate plans and targets among jurisdictions are compared, with and without recorded emissions inventories. This connects emissions measurement with formal climate planning and target-setting. Each jurisdiction is counted once, preventing those with multiple records from disproportionately influencing the percentages. Missing records do not necessarily mean no action exists.



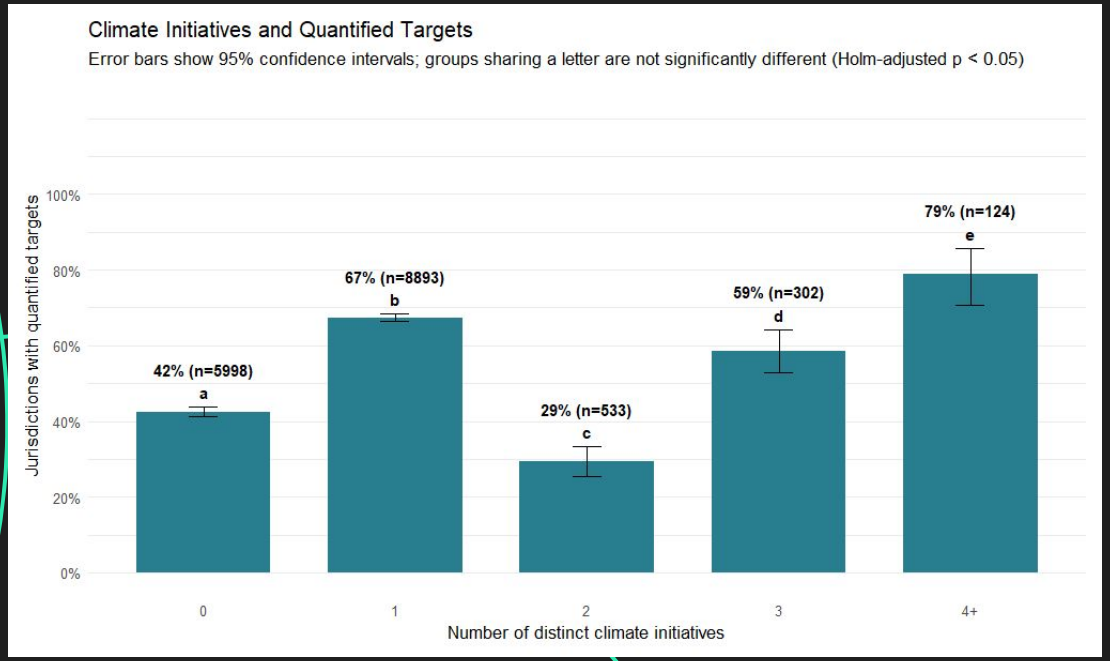
Why: Assess whether emissions measurement is associated with more formal climate planning

Data quality: Unrecorded plans or targets may be misclassified as absent

Updates: Reconcile jurisdiction IDs and track changes in reporting coverage

4. Are Climate Initiatives Associated With More Quantified Targets?

Examination of whether jurisdictions participating in more climate initiatives are more likely to have quantified targets. Grouping jurisdictions by initiative count makes it possible to compare the prevalence of measurable commitments across levels of engagement. A quantified target was defined using recorded numeric target values and units.



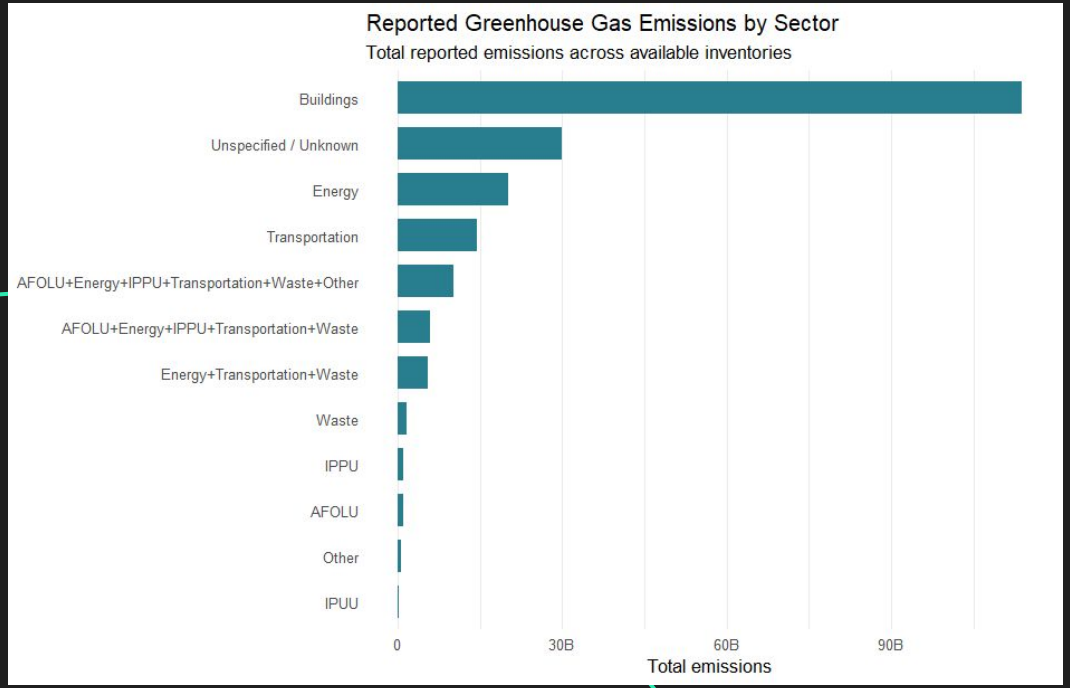
Why: Explore whether participation in more climate initiatives is associated with measurable commitments

Data quality: Initiative counts and target completeness depend on consistent records and units

Updates: De-duplicate initiatives and standardize target fields

Figure 5. Reported Greenhouse Gas Emissions by Sector.

Total reported emissions across available jurisdictional inventories, grouped by sector. Missing and unspecified sector classifications are combined into an “Unspecified / Unknown” category. Because the data include inventories from different years and jurisdictions, totals reflect the available records rather than a single-year global emissions estimate.



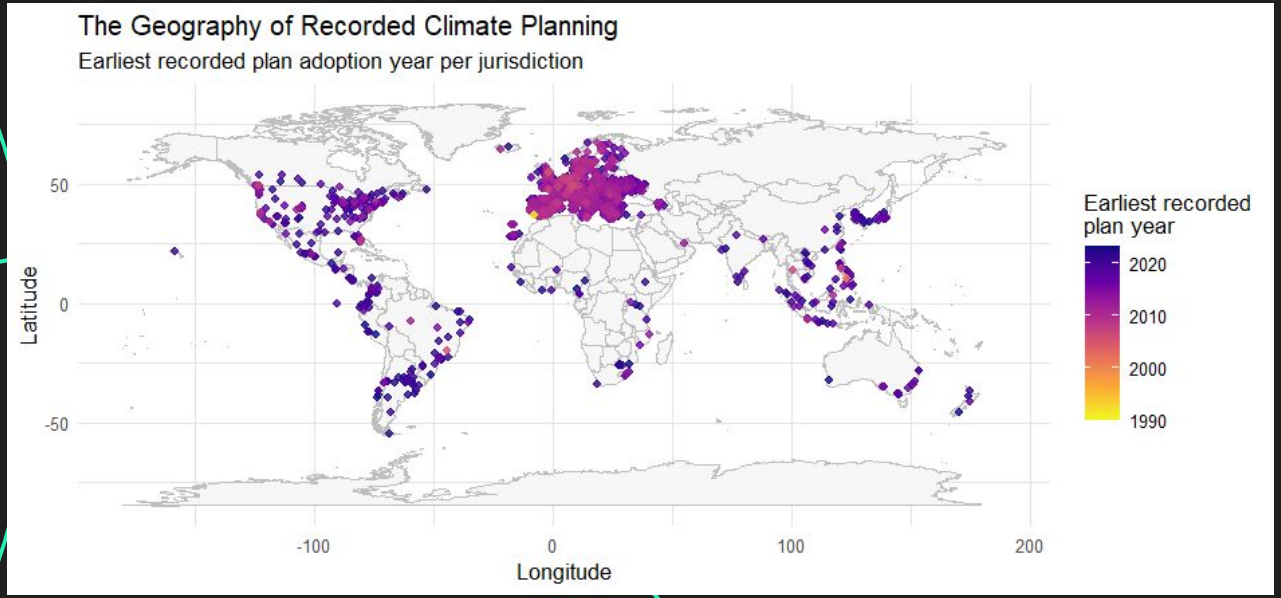
Why: Identify which sectors account for the largest share of reported emissions and may warrant attention

Data quality: Inventories cover different years; unspecified sectors limit interpretation

Updates: Harmonize sector classifications and compare consistent reporting periods

6. Where Was Climate Planning Recorded Earliest?

The earliest recorded climate-plan adoption year for each jurisdiction globally were mapped to explore the geographic and temporal spread of formal climate planning. The earliest year in the dataset may not be the jurisdiction's actual first plan. The map reflects available records, not a complete history of climate action.



Why: Explore when climate planning emerged geographically and how adoption spread over time

Data quality: Missing or inaccurate adoption dates and coordinates can distort patterns

Updates: Validate dates and locations and incorporate newly recorded plans